Exact and Heuristic Certification in Redistricting Algorithms

Gio Della-Libera

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Abstract

This draft separates exact certificates, heuristic evidence, audit sidecars, and legal interpretation. It explains what bounds, witnesses, and verifier contracts can establish, and where empirical or legal claims remain outside the certificate. Its central rule is that every certificate must name its model, instance, objective, verifier, and claim boundary.

1 Introduction

U.13 is a methodology and claim-discipline paper. It distinguishes exact optimization certificates from heuristic search evidence and from RPLAN/RCTX audit validation.

The distinction matters because the U-series deliberately mixes exact and heuristic work. Branch-and-cut and branch-and-price papers may eventually produce bounds or infeasibility witnesses for scoped formulations. LNS, evolutionary, ensemble, and Pareto workflows can produce useful plans, traces, and comparisons. Audit certificates can make either lineage inspectable. These outputs should meet at the fixed point without being described as the same kind of proof.

2 Certificate Classes

Exact certificates may include bounds, proofs of infeasibility, or solver reports for scoped instances. Heuristic evidence includes fixture correctness, repeatability, benchmark comparisons, and empirical sweeps. These evidence classes are useful but not interchangeable.

A certificate is therefore a sentence with required nouns: instance, model, objective, constraint profile, producing method, verifier, and result. Omitting one of these nouns turns a scoped technical claim into an ambiguous assertion.

3 Audit Artifacts

Audit artifacts verify declared profiles, lineage, parameters, manifests, and certificate payloads. They do not transform a heuristic result into an exact optimality proof or a legal

Artifact	Can support	Does not establish alone
Optimality bound	A scoped mathematical claim under a formulation	Fairness or legal sufficiency
Infeasibility witness	No solution under stated constraints	No acceptable plan under other models
Solver report	Reproducible run status and parameters	Correctness beyond trusted solver/model scope
Heuristic trace	Validity-preserving search behavior	Global optimality
Benchmark/sweep	Empirical comparison under protocol	Universal dominance
Audit certificate	Declared profile and lineage checks	Normative or legal conclusion

Table 1: Certificate and evidence classes used by exact and heuristic papers.

conclusion.

At the RPLAN/RCTX fixed point, the verifier should be able to answer three questions: what plan or certificate is being checked, which declared profile is being applied, and which upstream run context produced the payload. This is an audit contract, not a substitution for the mathematical or empirical evidence named by the payload itself.

4 Current Implementation Boundary

The current U-track implementation family now includes exact-method substrates (`bisect-ilp`, `bisect-column`), heuristic and frontier substrates (`bisect-local-search`, `bisect-pareto`, `bisect-ensemble`, `bisect-multiscale`), and verifier-facing RPLAN/RCTX packages. Certification language should follow the artifact: solver proof, bound report, heuristic diagnostic, selected-frontier package, or final-plan audit certificate.

The key rule remains unchanged but is now more operational: never let a heuristic diagnostic masquerade as a proof, and never let an audit certificate for declared context become a claim that the plan is legally sufficient in the world.

5 Limitations

Certificates are always scoped to a model, dataset, objective, and profile. The paper must avoid implying that exactness for one mathematical formulation settles broader legal, representational, or political questions.

The current implementation papers are staged at different evidence levels. Until U.16 and U.17 document solver contracts and fixtures, exact certificates should be described as target contracts. Until U.18 and U.19 document empirical protocols, heuristic claims should be described as validity and comparison claims rather than dominance claims.

6 Conclusion

U.13 gives the U-series vocabulary for saying what an algorithm has proved, what it has measured, and what it has merely recorded for audit.

That vocabulary is the safeguard: exact methods may make scoped proof claims, heuristics may make empirical and traceability claims, and audit certificates may make verification claims. Keeping those claims separate lets the portfolio be ambitious without becoming vague.

References

BISECT Project. U.13 exact-vs-heuristic certification plan, 2026. BISECT internal paper plan.